

Ameerbinmahfooz

RTL Design and Verification

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LinkedIn | GitHub



CAREER OBJECTIVE

Eager to contribute to innovative semiconductor projects by leveraging a strong foundation in Verilog-based RTL design and hands-on expertise with tools like VCS, Questa, Vivado, and Icarus Verilog for synthesis, implementation, and UVM-based verification. Analytical and collaborative, I am committed to continuous learning and passionate about delivering high-quality results. Driven to make a meaningful impact through efficient problem-solving and proactive communication.

PROFESSIONAL TRAINING

Maven Silicon Softech Private Limited

July 2024 – Dec 2025

Advanced VLSI Design and Verification

EDUCATION

Visvesvaraya Technological University (VTU), Bhatkal, Karnataka

April 2020 – May 2024

Bachelor of Engineering in Electronics and Communication

CGPA: 7.26/10

Anjuman PU College, Bhatkal, Karnataka

April 2018 – April 2020

PCMB (Class XII)

Percentage: 81%

Anjuman Boy's High School, Bhatkal, Karnataka

April 2017 – April 2018

Class X

Percentage: 91%

TECHNICAL SKILLS

HDL & Verification: Verilog, SystemVerilog, UVM

Tools & Platforms: Vivado, Synopsys VCS, Questasim, GTKWave, Ngspice, Iverilog, Git, Linux

Concepts: RTL Design, FPGA, OOP, FSM, STA

PROJECTS

AHB to APB Bridge Verification IP (VIP) | [Link](#)

- **Tools & Languages:** SystemVerilog, UVM, VCS, Questasim, Makefile
- **Verification IP Development:** Designed a reusable UVM-based VIP to verify the AHB-to-APB bridge, including master/slave agents, monitor, driver, scoreboard, and coverage.
- **Test Scenarios & Coverage:** Developed self-checking tests for read/write transactions, wait states, and error responses to ensure protocol compliance and functional coverage.

MiniCPU Design and Testbench Development | [Link](#)

- **Tools & Languages:** Verilog, Icarus Verilog, ModelSim, Vivado, Makefile
- **Modular 8-bit CPU:** Designed an 8-bit CPU with ALU, register bank, control unit, input/output ports, and instruction set supporting arithmetic, data transfer, and conditional jumps.
- **Self-Checking Testbench:** Developed a reusable, self-checking testbench with tasks for reset, instruction execution, pass/fail reporting, and functional coverage of all instruction types.

UVM-Based Router Design and Verification | [Link](#)

- **Tools & Languages:** Verilog, SystemVerilog, UVM, VCS, Questasim, Makefile
- **Router Design:** Designed packet-router with one input and three FIFO-buffered outputs using Verilog.

- **Verification:** Developed a UVM-based verification environment with agents, drivers, monitors, and a scoreboard for 100% functional coverage.

Medicine Box Disabled Person | Final Year Project | [Link](#)

- **Tools & Languages:** Arduino IDE, C++, ESP32
- **Cost-Effective Design:** Achieved a 10x cost reduction compared to market alternatives.
- **Market-Ready Innovation:** Developed a scalable, user-friendly product for healthcare.

CERTIFICATIONS

- Digital Systems: From Logic Gates to Processors - **Coursera**
- Hardware Modeling using Verilog - **NPTEL**
- Hands-on Introduction to Linux Commands and Shell Scripting - **Coursera**
- Machine Learning with Python (with Honors) - **Coursera**
- Electronic Device and Characterization - **IIT Bombay**

ACHIEVEMENTS

- Conducted two technical workshops on Arduino, training 30+ participants in micro-controller programming and project development.
- Organized a district-level quiz competition, attracting 240+ participants and gaining recognition for fostering technical engagement.
- Gold Medalist at Anjuman Boys High School for academic, sports and literacy excellence.

STRENGTHS

- Problem-Solving
- Quick Learner
- Team Player

HOBBIES

- Chess
- Football

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I take full responsibility for the accuracy of the particulars mentioned in this resume.

Date:

Place: Bangalore, Karnataka

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